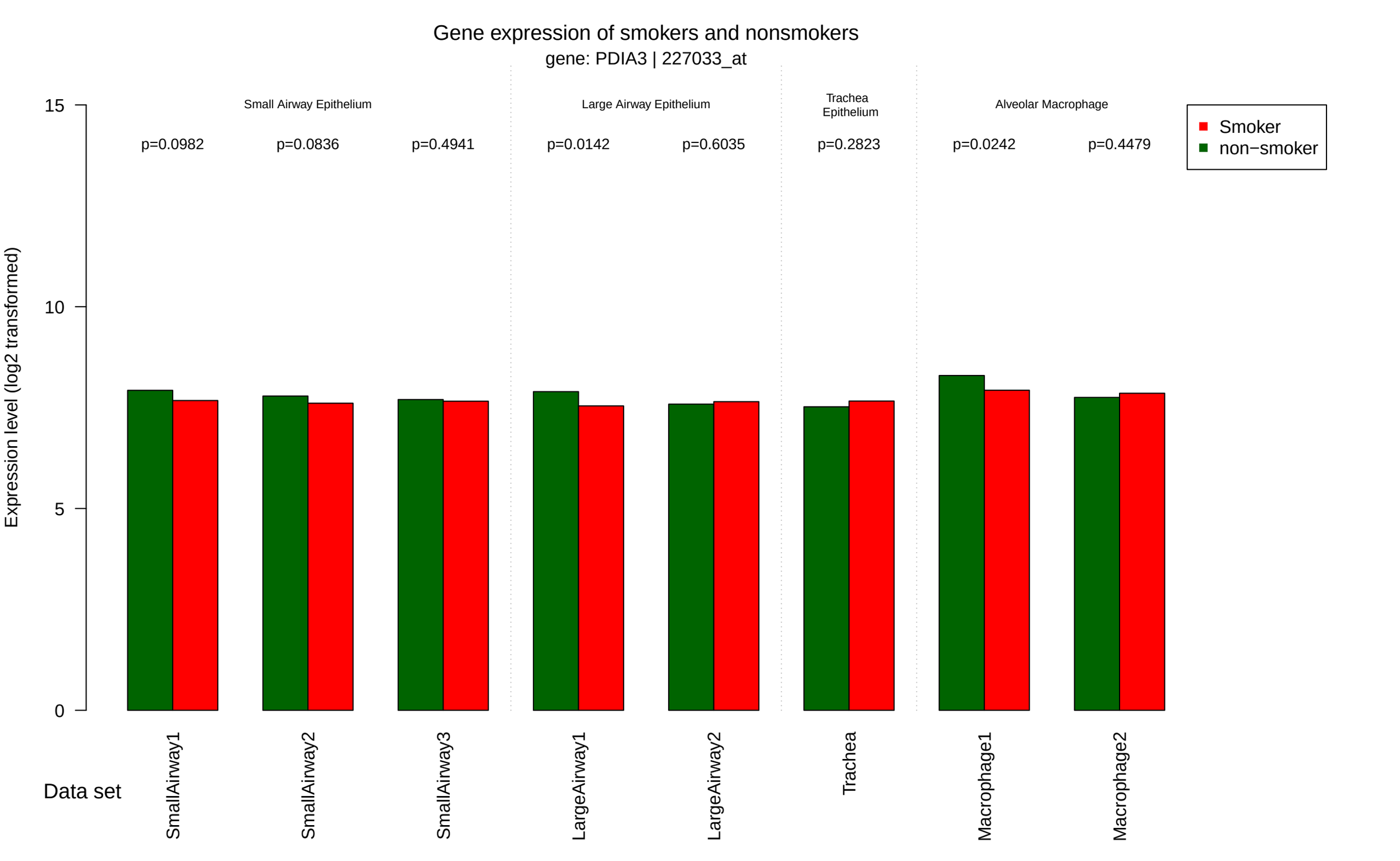
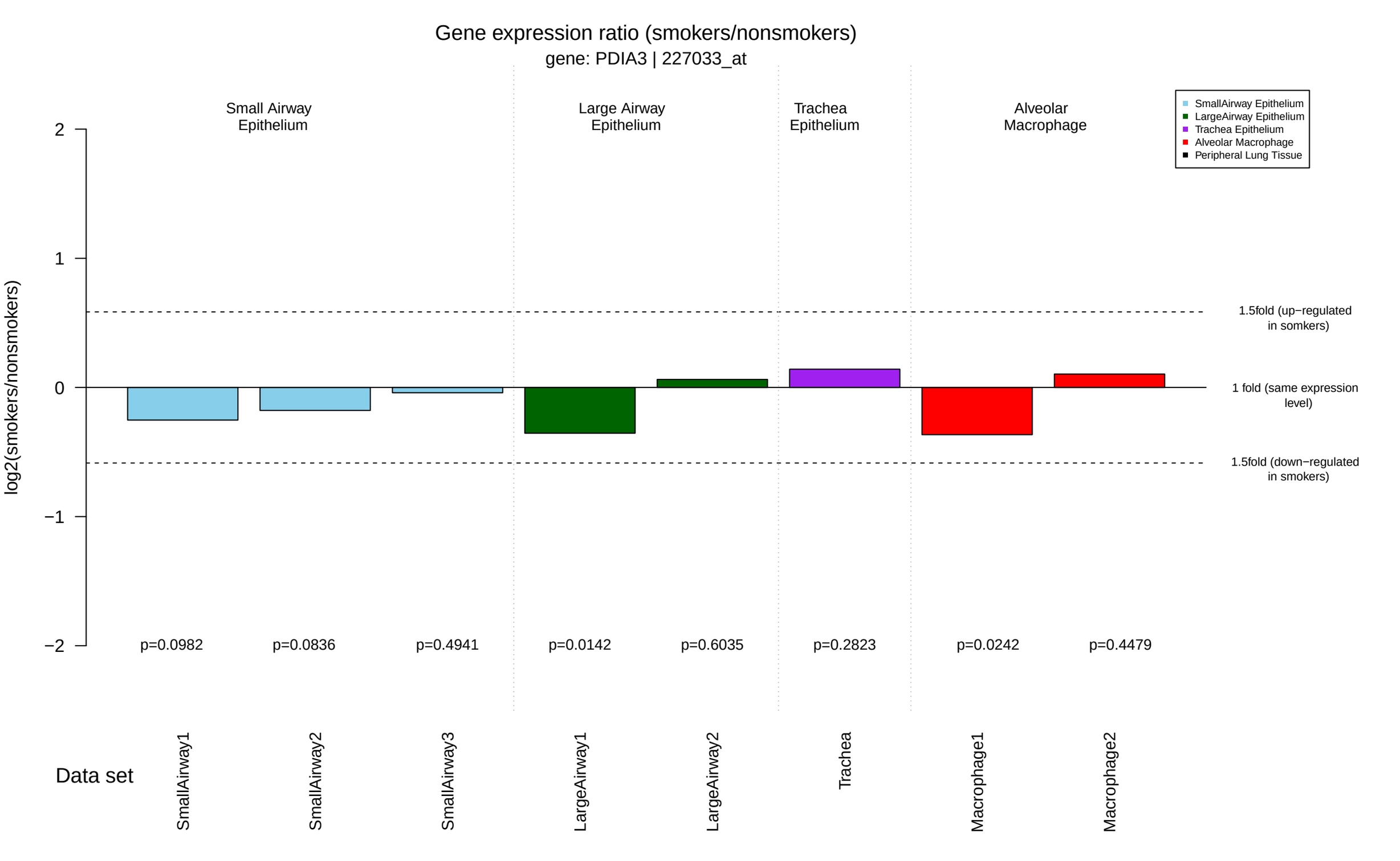
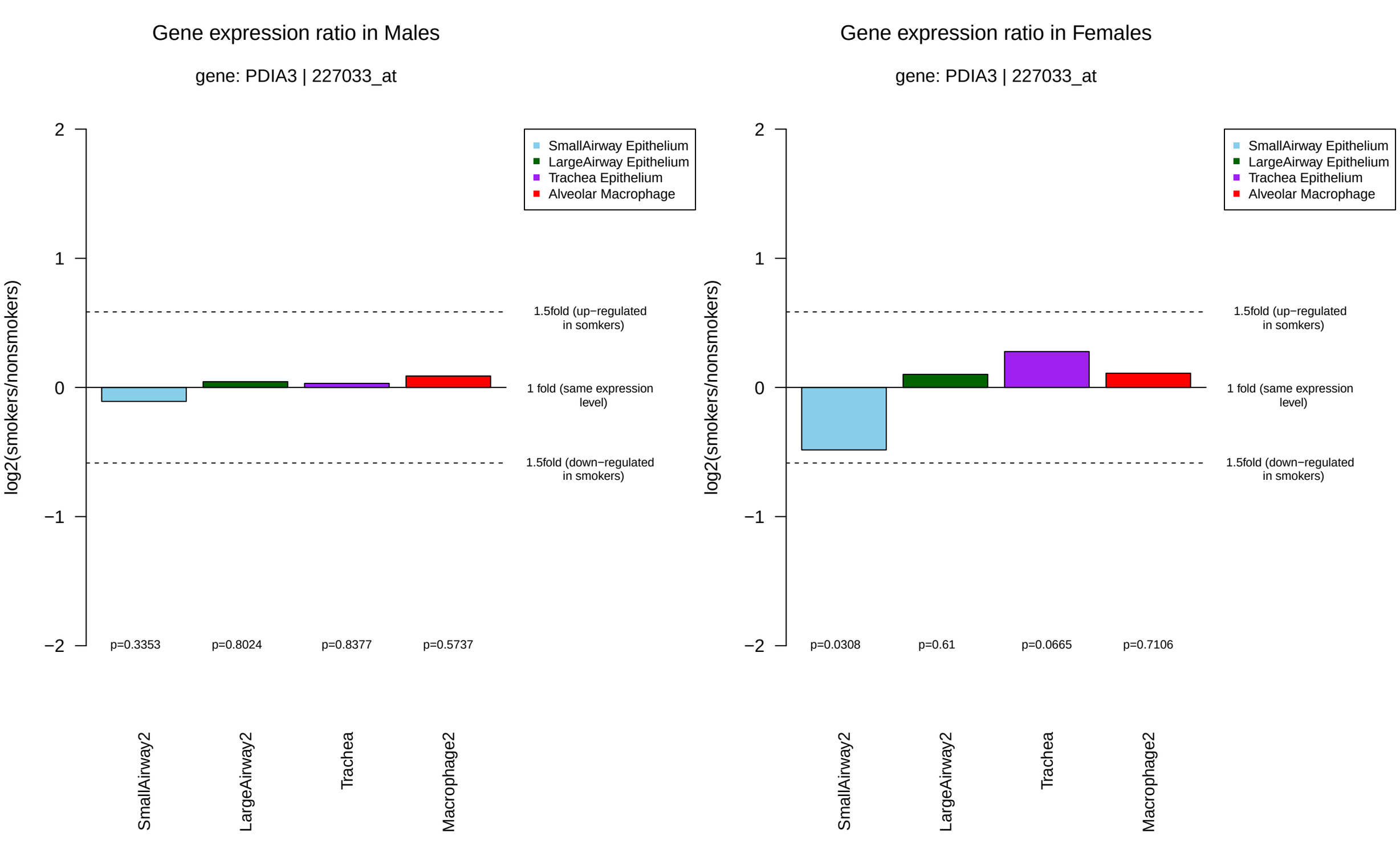
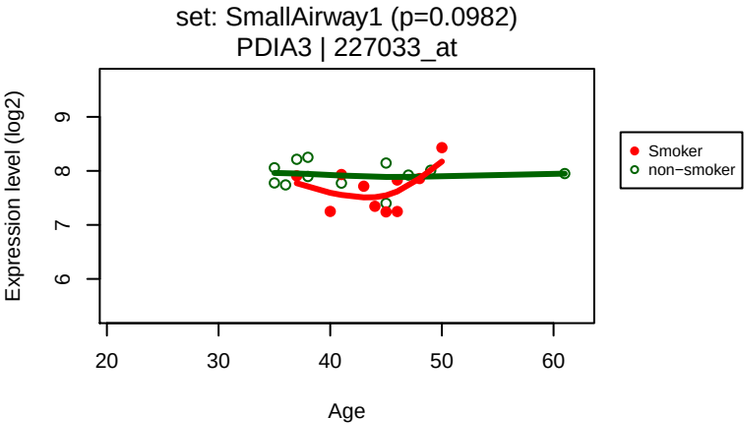
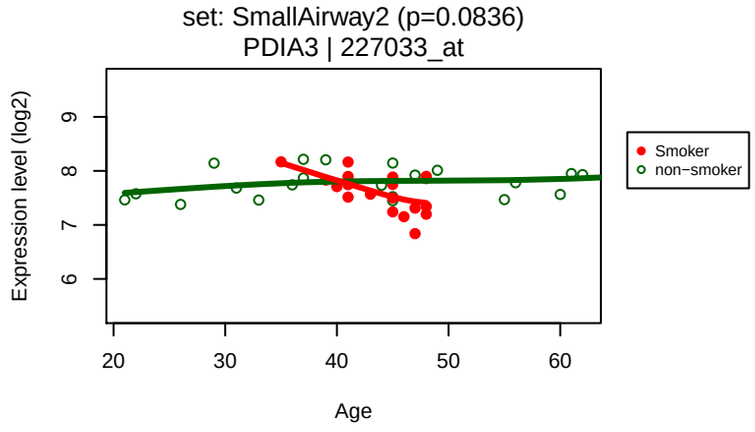
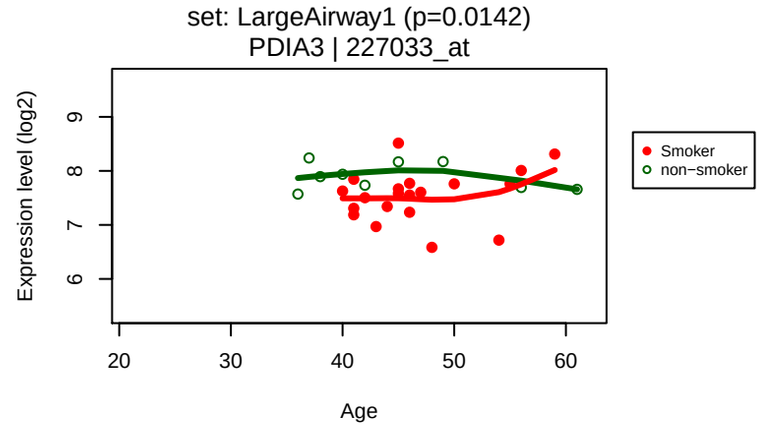

Gene expression of smokers and nonsmokers
gene: PDIA3 | 227033_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0982
p=0.0836
p=0.4941
p=0.0142
p=0.6035
p=0.2823
p=0.0242
p=0.4479
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

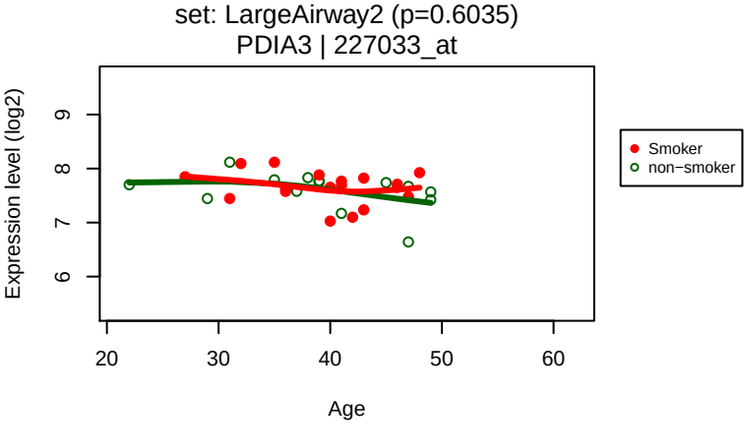

Gene expression ratio (smokers/nonsmokers)
gene: PDIA3 | 227033_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0982
p=0.0836
p=0.4941
p=0.0142
p=0.6035
p=0.2823
p=0.0242
p=0.4479
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

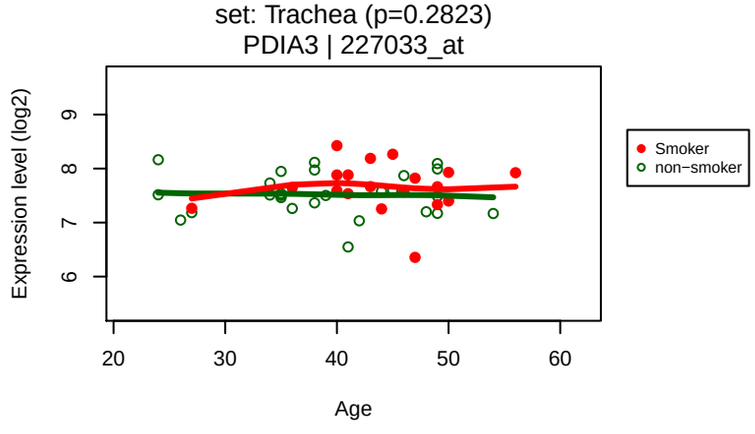

Gene expression ratio in Males
gene: PDIA3 | 227033_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3353
p=0.8024
p=0.8377
p=0.5737
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PDIA3 | 227033_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0308
p=0.61
p=0.0665
p=0.7106
SmallAirway2
LargeAirway2
Trachea
Macrophage2

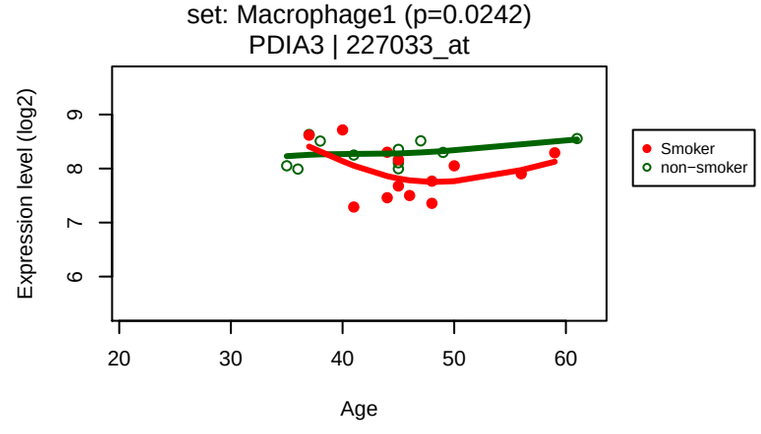

set: SmallAirway1 (p=0.0982)
PDIA3 | 227033_at
Expression level (log2)
Age
Smoker
non−smoker

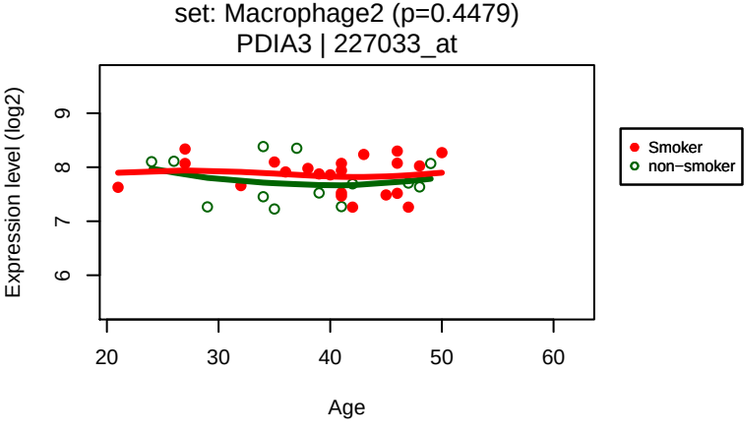

set: SmallAirway2 (p=0.0836)
PDIA3 | 227033_at
Expression level (log2)
Age
Smoker
non−smoker

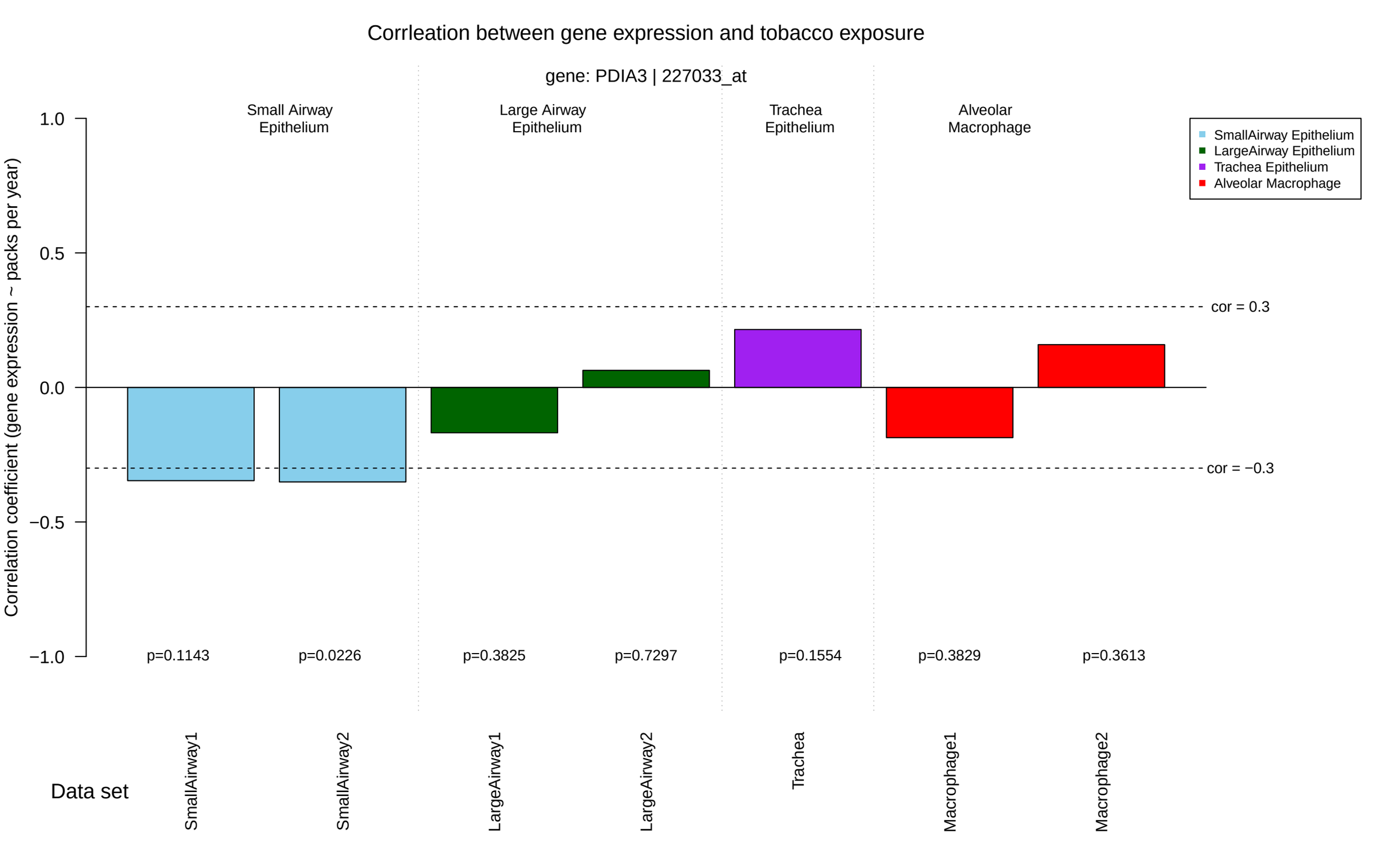

set: LargeAirway1 (p=0.0142)
PDIA3 | 227033_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6035)
PDIA3 | 227033_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2823)
PDIA3 | 227033_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0242)
PDIA3 | 227033_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4479)
PDIA3 | 227033_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PDIA3 | 227033_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1143
p=0.0226
p=0.3825
p=0.7297
p=0.1554
p=0.3829
p=0.3613
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2